

Introduction to Artificial Intelligence

The Turing Test and the Question of Machine Intelligence; Human vs Machine Intelligence

The Turing Test

Proposed by Alan Turing in his 1950 paper "Computing Machinery and Intelligence," the Turing Test (originally called the "Imitation Game") was designed to sidestep the philosophically difficult question "Can machines think?" and replace it with an operational, testable question.

How the Test Works

- A human judge holds text-based conversations with two hidden participants: one human, one machine.
- The judge does not know which is which and must decide, based only on the conversation, which participant is the machine.
- If the judge cannot reliably distinguish the machine from the human, the machine is said to have "passed" the test.

Significance

- Shifted the debate on machine intelligence from an abstract philosophical question to an observable, behavioural criterion.
 - Became a foundational benchmark and cultural touchstone for AI, referenced in nearly every discussion of machine intelligence since.
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Criticisms of the Turing Test

- **The Chinese Room Argument (John Searle, 1980):** Searle argued that a system could manipulate symbols to produce convincing responses (passing the test) without any real understanding — the machine merely follows rules, like a person in a room manually following instructions to convert Chinese input into Chinese output, without understanding Chinese.
 - The test measures the ability to imitate human conversation, not genuine intelligence, understanding, or consciousness.
 - It is anthropocentric — it assumes human-like conversation is the correct benchmark for intelligence, which may exclude other valid forms of intelligent behaviour.
 - Modern chatbots can sometimes fool judges through clever tricks (evasive answers, typos, humour) rather than true reasoning.
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Human Intelligence vs Machine Intelligence

Aspect	Human Intelligence	Machine Intelligence
Basis	Biological neural networks, consciousness, lived experience	Algorithms, data, mathematical/statistical models
Learning	Contextual, intuitive; learns from very few examples	Often needs large volumes of labelled data
Common Sense	Strong innate common-sense reasoning	Limited; struggles with implicit/contextual knowledge
Creativity & Emotion	Genuine emotion, subjective creativity, empathy	Simulated patterns; no subjective experience
Consistency	Can be inconsistent, fatigued, biased by mood	Highly consistent and tireless within its trained scope
Speed & Scale	Limited processing speed and memory	Extremely fast computation and large-scale data processing
Adaptability to novelty	Can generalise well to entirely new situations	Generalises poorly outside its training distribution

The comparison shows that human and machine intelligence are complementary rather than directly substitutable: machines excel at speed, scale, and consistency, while humans excel at contextual understanding, common sense, and genuine creativity.

Symbolic (Rule-Based) AI vs Data-Driven (Learning-Based) AI; Hierarchy and Taxonomy of AI, ML, DL and Generative AI

Symbolic (Rule-Based) AI

Symbolic AI, also called "Good Old-Fashioned AI" (GOFAI), represents knowledge using explicit symbols and rules, and derives conclusions through logical inference.

- Knowledge is hand-coded by human experts as IF–THEN rules or logical statements.
- Reasoning is performed by an "inference engine" that applies rules to known facts to derive new facts.
- Example: An expert system for medical diagnosis such as MYCIN, which used rules like "IF the organism is gram-negative AND the site is blood, THEN consider organism X."
- Strengths: Transparent, explainable, precise within a well-defined domain.
- Weaknesses: Brittle — struggles with ambiguity, exceptions, and situations not explicitly coded; scaling requires enormous manual effort.

Data-Driven (Learning-Based) AI

Data-driven AI derives patterns and rules automatically from large volumes of data, rather than relying on manually coded rules.

- Systems are "trained" on data; statistical/mathematical models adjust internal parameters to minimise prediction error.
- Example: A spam filter that learns to classify emails by analysing thousands of labelled examples, rather than following explicit hand-written spam rules.
- Strengths: Handles ambiguity and complexity well; improves as more data becomes available; generalises to unseen patterns.
- Weaknesses: Requires large datasets; can be a "black box" (hard to explain decisions); can inherit and amplify biases present in training data.

Symbolic vs Data-Driven AI — Comparison

Feature	Symbolic (Rule-Based) AI	Data-Driven (Learning-Based) AI
Knowledge source	Explicit rules from human experts	Patterns learned automatically from data
Transparency	High — decisions are traceable to rules	Often low — "black box" behaviour
Adaptability	Low — must be manually updated	High — improves with more/new data
Data requirement	Minimal	Large volumes typically required
Example application	Expert systems, logic-based planners	Image recognition, recommendation systems

Hierarchy and Taxonomy: AI, ML, Deep Learning and Generative AI

These four terms are often used loosely but represent a nested hierarchy, each being a subset of the one before it:

- **Artificial Intelligence (outermost):** The broad field of building machines that exhibit intelligent behaviour, including both symbolic and data-driven approaches.
- **Machine Learning (subset of AI):** A data-driven approach where systems learn patterns from data rather than being explicitly programmed with rules.
- **Deep Learning (subset of ML):** A family of ML techniques based on multi-layered artificial neural networks, capable of automatically learning hierarchical features from raw data (e.g., images, audio, text).
- **Generative AI (subset of Deep Learning):** Deep learning models specifically designed to generate new content — text, images, audio, code — that resembles the data they were trained on (e.g., large language models, image generators).

Visualise this as four concentric circles: the largest circle is AI; inside it sits ML; inside ML sits Deep Learning; and inside Deep Learning sits the newest, most specialised circle — Generative AI.